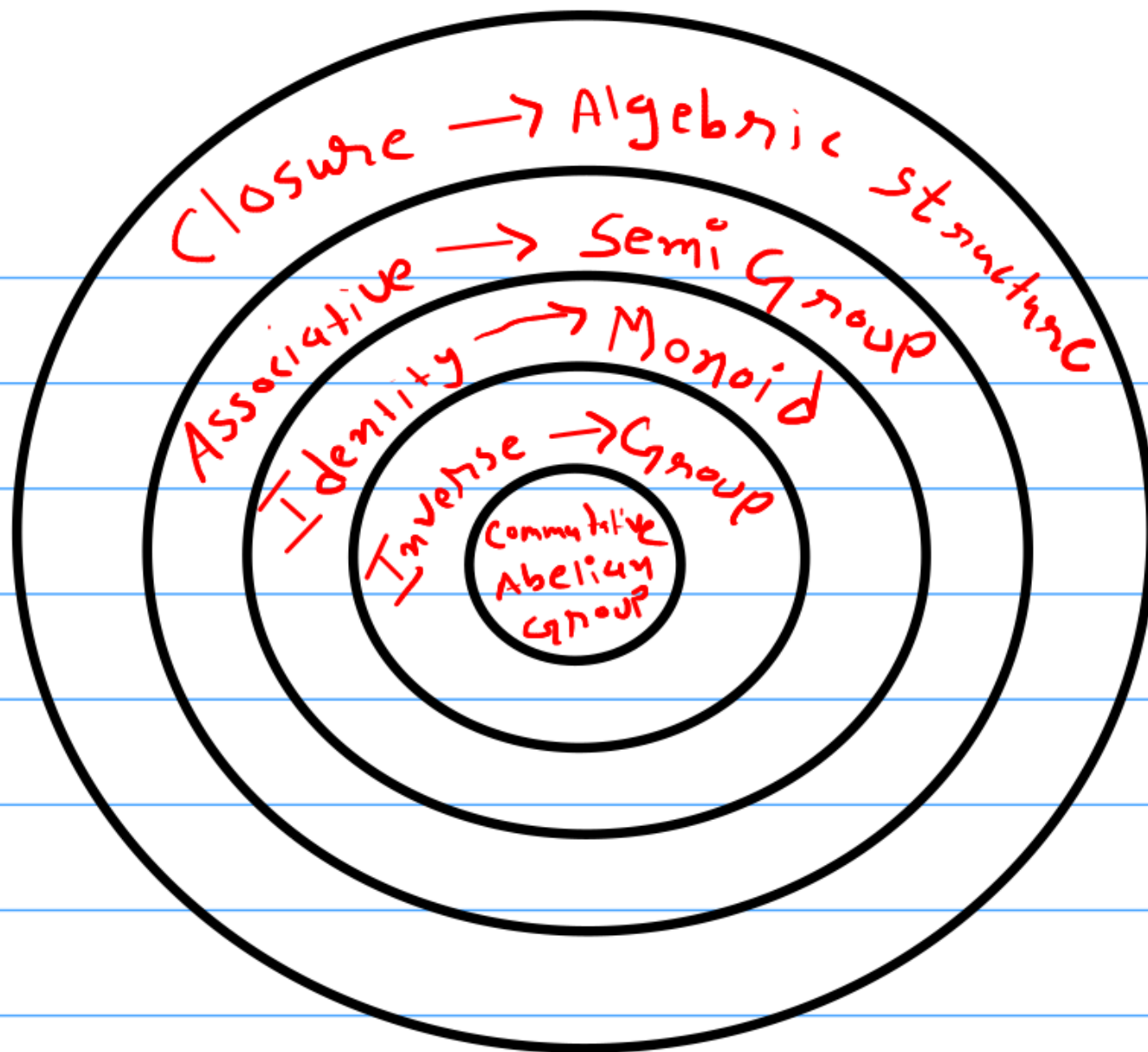


Group Theory

Introduction

- Set theory : Study of set operations.
- Relations : Study of pairs generated by product of two sets.
- Group Theory : Study of set with respect to some operation.



Algebraic Structure

- A non empty set S is called an algebraic structure w.r.t binary operation $(*)$ if it follows following axioms:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.

Let $S = \{-1, 1\}$

is $(S, *)$ algebraic structure?

$$-1 \times -1 = 1$$

$$-1 \times 1 = -1$$

$$1 \times 1 = 1$$

$$1 \times -1 = -1$$

$\in S$

So, YES

Commonly used sets

- **Natural Numbers (N)**, (also called positive integers, counting numbers, or natural numbers); They are the numbers $\{1, 2, 3, 4, 5, \dots\}$
- **Integers (Z)**. This is the set of all whole numbers plus all the negatives (or opposites) of the natural numbers, i.e., $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- **Rational numbers (Q)**. This is all the fractions where the top and bottom numbers are integers; e.g., $1/2, 3/4, 7/2, -4/3, 4/1$ [Note: The denominator cannot be 0, but the numerator can be].
- **Real numbers (R)**, (also called measuring numbers or measurement numbers). This includes all numbers that can be written as a decimal. This includes fractions written in decimal form e.g., 0.5, 0.75 2.35, -0.073 , 0.3333, or 2.142857. It also includes all the irrational numbers such as π , $\sqrt{2}$ etc. Every real number corresponds to a point on the number line.

Algebraic Structure Examples

	Algebraic System
$\mathbb{N}, +$	
$\mathbb{N}, -$	
$\mathbb{N}, *$	
$\mathbb{N}, /$	
$\mathbb{Z}, +$	
$\mathbb{Z}, -$	
$\mathbb{Z}, *$	
$\mathbb{Z}, /$	
$\mathbb{R}, +$	
$\mathbb{R}, -$	
$\mathbb{R}, *$	
$\mathbb{R}, /$	
$\mathbb{E}, +$	
$\mathbb{E}, *$	
$\mathbb{O}, +$	
$\mathbb{O}, *$	
$\mathbb{M}, +$	
$\mathbb{M}, *$	

	Algebraic System
$N, +$	Yes
$N, -$	No
$N, *$	Yes
$N, /$	No
$Z, +$	Yes
$Z, -$	Yes
$Z, *$	Yes
$Z, /$	No
$R, +$	Yes
$R, -$	Yes
$R, *$	Yes
$R, /$	No
$E, +$	Yes
$E, *$	Yes
$O, +$	No
$O, *$	Yes
$M, +$	Yes
$M, *$	Yes

Semigroup

- A non-empty set S , $(S, *)$ is called a semigroup if it follows the following axiom:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
 - **Associativity:** $a*(b*c) = (a*b)*c \quad \forall a, b, c$ belongs to S .

	Algebraic System	Semigroup
$\mathbb{N}, +$	Yes	
$\mathbb{N}, -$	No	
$\mathbb{N}, *$	Yes	
$\mathbb{N}, /$	No	
$\mathbb{Z}, +$	Yes	
$\mathbb{Z}, -$	Yes	
$\mathbb{Z}, *$	Yes	
$\mathbb{Z}, /$	No	
$\mathbb{R}, +$	Yes	
$\mathbb{R}, -$	Yes	
$\mathbb{R}, *$	Yes	
$\mathbb{R}, /$	No	
$\mathbb{E}, +$	Yes	
$\mathbb{E}, *$	Yes	
$\mathbb{O}, +$	No	
$\mathbb{O}, *$	Yes	
$\mathbb{M}, +$	Yes	
$\mathbb{M}, *$	Yes	

	Algebraic System	Semigroup
$\mathbb{N}, +$	Yes	Yes
$\mathbb{N}, -$	No	No
$\mathbb{N}, *$	Yes	Yes
$\mathbb{N}, /$	No	No
$\mathbb{Z}, +$	Yes	Yes
$\mathbb{Z}, -$	Yes	No
$\mathbb{Z}, *$	Yes	Yes
$\mathbb{Z}, /$	No	No
$\mathbb{R}, +$	Yes	Yes
$\mathbb{R}, -$	Yes	No
$\mathbb{R}, *$	Yes	Yes
$\mathbb{R}, /$	No	No
$\mathbb{E}, +$	Yes	Yes
$\mathbb{E}, *$	Yes	Yes
$\mathbb{O}, +$	No	No
$\mathbb{O}, *$	Yes	Yes
$\mathbb{M}, +$	Yes	Yes
$\mathbb{M}, *$	Yes	Yes

Monoid

- A non-empty set S , $(S, *)$ is called a monoid if it follows the following axiom:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
 - **Associativity:** $a*(b*c) = (a*b)*c \quad \forall a, b, c$ belongs to S .
 - **Identity Element:** There exists $e \in S$ such that $a*e = e*a = a \quad \forall a \in S$

	Algebraic System	Semigroup	Monoid
$\mathbb{N}, +$	Yes	Yes	
$\mathbb{N}, -$	No	No	
$\mathbb{N}, *$	Yes	Yes	
$\mathbb{N}, /$	No	No	
$\mathbb{Z}, +$	Yes	Yes	
$\mathbb{Z}, -$	Yes	No	
$\mathbb{Z}, *$	Yes	Yes	
$\mathbb{Z}, /$	No	No	
$\mathbb{R}, +$	Yes	Yes	
$\mathbb{R}, -$	Yes	No	
$\mathbb{R}, *$	Yes	Yes	
$\mathbb{R}, /$	No	No	
$\mathbb{E}, +$	Yes	Yes	
$\mathbb{E}, *$	Yes	Yes	
$\mathbb{O}, +$	No	No	
$\mathbb{O}, *$	Yes	Yes	
$\mathbb{M}, +$	Yes	Yes	
$\mathbb{M}, *$	Yes	Yes	

	Algebraic System	Semigroup	Monoid
$\mathbb{N}, +$	Yes	Yes	No
$\mathbb{N}, -$	No	No	No
$\mathbb{N}, *$	Yes	Yes	Yes
$\mathbb{N}, /$	No	No	No
$\mathbb{Z}, +$	Yes	Yes	Yes
$\mathbb{Z}, -$	Yes	No	No
$\mathbb{Z}, *$	Yes	Yes	Yes
$\mathbb{Z}, /$	No	No	No
$\mathbb{R}, +$	Yes	Yes	Yes
$\mathbb{R}, -$	Yes	No	No
$\mathbb{R}, *$	Yes	Yes	Yes
$\mathbb{R}, /$	No	No	No
$\mathbb{E}, +$	Yes	Yes	Yes
$\mathbb{E}, *$	Yes	Yes	No
$\mathbb{O}, +$	No	No	No
$\mathbb{O}, *$	Yes	Yes	Yes
$\mathbb{M}, +$	Yes	Yes	Yes
$\mathbb{M}, *$	Yes	Yes	Yes

Group

- A non-empty set G , $(G, *)$ is called a group if it follows the following axiom:
 - **Closure:** $(a*b)$ belongs to G for all $a, b \in G$.
 - **Associativity:** $a*(b*c) = (a*b)*c \quad \forall a, b, c$ belongs to G .
 - **Identity Element:** There exists $e \in G$ such that $a*e = e*a = a \quad \forall a \in G$
 - **Inverses:** $\forall a \in G$ there exists $a^{-1} \in G$ such that $a*a^{-1} = a^{-1}*a = e$

	Algebraic System	Semigroup	Monoid	Group
$\mathbb{N}, +$	Yes	Yes	No	
$\mathbb{N}, -$	No	No	No	
$\mathbb{N}, *$	Yes	Yes	Yes	
$\mathbb{N}, /$	No	No	No	
$\mathbb{Z}, +$	Yes	Yes	Yes	
$\mathbb{Z}, -$	Yes	No	No	
$\mathbb{Z}, *$	Yes	Yes	Yes	
$\mathbb{Z}, /$	No	No	No	
$\mathbb{R}, +$	Yes	Yes	Yes	
$\mathbb{R}, -$	Yes	No	No	
$\mathbb{R}, *$	Yes	Yes	Yes	
$\mathbb{R}, /$	No	No	No	
$\mathbb{E}, +$	Yes	Yes	Yes	
$\mathbb{E}, *$	Yes	Yes	No	
$\mathbb{O}, +$	No	No	No	
$\mathbb{O}, *$	Yes	Yes	Yes	
$\mathbb{M}, +$	Yes	Yes	Yes	
$\mathbb{M}, *$	Yes	Yes	Yes	

Abelian Group or Commutative group

- A non-empty set S , $(S, *)$ is called a Abelian group if it follows the following axiom:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
 - **Associativity:** $a*(b*c) = (a*b)*c \quad \forall a, b, c$ belongs to S .
 - **Identity Element:** There exists $e \in S$ such that $a*e = e*a = a \quad \forall a \in S$
 - **Inverses:** $\forall a \in S$ there exists $a^{-1} \in S$ such that $a*a^{-1} = a^{-1}*a = e$
 - **Commutative:** $a*b = b*a$ for all $a, b \in S$

	Algebraic System	Semigroup	Monoid	Group	Abelian Group
$\mathbb{N}, +$	Yes	Yes	No	No	
$\mathbb{N}, -$	No	No	No	No	
$\mathbb{N}, *$	Yes	Yes	Yes	No	
$\mathbb{N}, /$	No	No	No	No	
$\mathbb{Z}, +$	Yes	Yes	Yes	Yes	
$\mathbb{Z}, -$	Yes	No	No	No	
$\mathbb{Z}, *$	Yes	Yes	Yes	No	
$\mathbb{Z}, /$	No	No	No	No	
$\mathbb{R}, +$	Yes	Yes	Yes	Yes	
$\mathbb{R}, -$	Yes	No	No	No	
$\mathbb{R}, *$	Yes	Yes	Yes	No	
$\mathbb{R}, /$	No	No	No	No	
$\mathbb{E}, +$	Yes	Yes	Yes	Yes	
$\mathbb{E}, *$	Yes	Yes	No	No	
$\mathbb{O}, +$	No	No	No	No	
$\mathbb{O}, *$	Yes	Yes	Yes	No	
$\mathbb{M}, +$	Yes	Yes	Yes	Yes	
$\mathbb{M}, *$	Yes	Yes	Yes	No	

	Algebraic System	Semigroup	Monoid	Group	Abelian Group
$\mathbb{N}, +$	Yes	Yes	No	No	No
$\mathbb{N}, -$	No	No	No	No	No
$\mathbb{N}, *$	Yes	Yes	Yes	No	No
$\mathbb{N}, /$	No	No	No	No	No
$\mathbb{Z}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{Z}, -$	Yes	No	No	No	No
$\mathbb{Z}, *$	Yes	Yes	Yes	No	No
$\mathbb{Z}, /$	No	No	No	No	No
$\mathbb{R}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{R}, -$	Yes	No	No	No	No
$\mathbb{R}, *$	Yes	Yes	Yes	No	No
$\mathbb{R}, /$	No	No	No	No	No
$\mathbb{E}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{E}, *$	Yes	Yes	No	No	No
$\mathbb{O}, +$	No	No	No	No	No
$\mathbb{O}, *$	Yes	Yes	Yes	No	No
$\mathbb{M}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{M}, *$	Yes	Yes	Yes	No	No

- $(\mathbb{N}, *)$ where $\mathbb{N}$ is natural number. $*$ refers to the power operation.
 - It is semi group
 - It is not semi group
 - It is monoid but not group
 - It is Group

- $(N, *)$ where N is natural number. $*$ refers to the power operation.
 - It is semi group
 - **It is not semi group**
 - It is monoid but not group
 - It is Group

- $(Z, *)$ where Z is natural number. $*$ refers to the Max operation.
 - Algebraic structure ? Semigroup ? Group ?
- $(Z, *)$ where Z is natural number. $*$ refers to the Min operation.
 - Algebraic structure ? Semigroup ? Group ?

- $(Z, *)$ where Z is natural number. $*$ refers to the Max operation.
 - Algebraic structure ? **Semigroup** ? Group ?
- $(Z, *)$ where Z is natural number. $*$ refers to the Min operation.
 - Algebraic structure ? **Semigroup** ? Group ?

~ Let $Q^+ =$ Set of Positive Rational No.
 $a * b = \frac{a \cdot b}{3}$, $(Q^+, *)$ is abelian Group.

a) $e = 3$

b) $a^{-1} = \frac{9}{a}$

3) $(\frac{2}{3})^{-1} = 6$

4) $3^+ = 3$

- Which of the following is not a group?

a) $\{0, \pm 2, \pm 4, \pm 6, \dots, \infty\}, +$

b) $\{0, \pm k, \pm 2k, \pm 3k, \dots, \infty\}, +$ (k is Natural No.)

c) $\{2^n \mid n \text{ is integer}\}, *$

d) $\{\text{Set of all Complex no}, *\}$